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Task Description:	A unique feature of the space radiation environment is the presence of high-energy charged particles including protons and fully ionized atomic nuclei such as 56Fe. These particulate radiations may pose a significant hazard to space flight crews not only during the mission but also later, on Earth, when slow-developing adverse effects finally become apparent. In the CNS, exposure to such radiation may not only directly affect structure/function within the brain, but it may also make tissues more sensitive to a secondary insult such as trauma, stoke or degenerative disease. We have shown that photon and particulate irradiations induce a strong and persistent inflammatory response in the brain that is associated with altered hippocampal neurogenesis and impaired cognitive function. Based on these data we hypothesize that such inflammatory changes, while they may only lead to minor morphologic/functional changes by themselves, will predispose the brain, and in particular hippocampal neurogenesis, to a heightened vulnerability after a secondary insult. Secondly, we hypothesize that a pharmacologic reduction of radiation-induced inflammation will ameliorate the combined effects of radiation and secondary injury on hippocampal neurogenesis. While the nature of possible secondary insults is variable, we have identified traumatic brain injury (TBI) as the choice for the proposed studies. We have identified 4 specific aims to test our hypotheses: 1) determine if inflammatory changes induced by a low dose of proton irradiation predisposes the hippocampus to abnormal neurogenesis after secondary TBI; 2) determine if inflammatory changes induced by a low dose of 56Fe irradiation predisposes the hippocampus to abnormal neurogenesis after secondary TBI; 3) using knock-out mice lacking chemokine receptor 2 (CCR2), determine if a muted monocytic inflammatory response can reverse the influence of low dose proton and/or 56Fe irradiation on TBI-induced alterations in neurogenesis; and 4) determine if pharmacologic intervention after proton and 56Fe irradiation will reduce radiation-induced inflammation and lessen the effects of subsequent TBI. Mice will be irradiated with low radiation doses (0–2 Gy) and 30 days later will subjected to mild/moderate TBI. Neurogenesis and inflammation will be detected using cell specific immunohistochemistry and the effects of irradiation will be quantified using confocal microscopy and unbiased stereology. These studies will address a possible mechanism by which low-dose irradiation may sensitize the brain to secondary injury, and they will also involve a pharmacologic approach to minimize such effects.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Neurogenesis, or the generation of new neurons, continues throughout life in the hippocampal formation of all mammals, including humans. Neurogenesis is associated with hippocampal dependent cognitive functions, and reduced neurogenesis is linked to poorer cognitive function. Radiation alone and traumatic brain injury (TBI) alone both reduce neurogenesis and are correlated with hippocampal dependent cognitive impairments. Both of these insults occur in the context of normal life on earth, either via therapeutic use (irradiation) or as a result of accident or injury (TBI). Given that TBI is a relatively common type of injury, especially in young adults, it is conceivable that individuals receiving therapeutic irradiation for treatment of brain tumors, head and neck tumors, etc., could experience combined injury. Additionally, given the state of the world at this time, combined injury in a nuclear battlefield or in an uncontrolled radiation exposure such as a dirty bomb, is a possible outcome of a terrorist event. As a consequence, understanding the mechanisms associated with radiation, TBI alone or combined injury is essential for the development of suitable and effective countermeasures or treatments.
	In the development of this research project we hypothesized that after irradiation with 56Fe ions, radiation-induced inflammatory changes in the brain, while they may only lead to minor morphologic/functional changes by themselves, would predispose the brain to heightened vulnerability after a secondary insult. We identified traumatic brain injury as the choice for the secondary insult because it is a common injury in adults and because we have shown it to induce a reproducible injury involving inflammation and impaired neurogenesis. Thus, the experiments performed under the auspices of this research involved the measurements of cellular, molecular and behavioral effects induced after sham treatment, radiation only, traumatic brain injury only and combined injury. While technical and practical issues (summarized in previous progress reports) impacted the ability to carry out all of the originally proposed experiments, we were able to perform many as originally designed. In addition, we were able to integrate techniques and approaches not originally proposed, that provided new data on specific molecular and behavioral changes. This task book summary addresses only the results not previously reported. During the last two years of the project we were able to integrate direct measures of cognition as well as measurement of the expression of an immediate early gene (IEG) that is associated with behavioral performance. Arc (activity-regulated cytoskeleton-associated protein) is rapidly induced in response to synaptic activity and is the only known activity-induced gene that correlates both temporally and spatially with the stimulus that induced its transcription. Furthermore, Arc is the most responsive IEG to differences in behavioral task demands, is essential for consolidation of synaptic plasticity and memory, and when compared to other IEGs, has a distinct role in modulating hippocampal synaptic plasticity. Thus, we assert that the determination of Arc expression is an appropriate approach to map neuronal function that underlies information processing and behavioral performance. Arc expression and cognition were combined with our well-established methods to quantify neurogenesis. Our behavioral testing involved the Morris water maze, a test that facilitated the measurement of learning and memory retention. There were no group differences in swim speeds during the visible platform sessions, indicating that there were no deficits in motor or exploratory behavior as result of single or combined injury. Additionally, all groups showed similar abilities to learn in the visible platform version of the water maze. Memory retention was assessed using a probe trial, where animals had to remember to position of a hidden platform (now removed) using spatial cues in the room within which the maze was situated. While mice that were sham treated or received radiation only showed spatial memory retention, mice that received trauma only did not. Mice that received trauma 30 days after 0.5 Gy of 56Fe, also showed spatial memory retention. These results suggested that trauma induced a significant effect on memory retention, but when radiation preceded it there was significant recovery. This implied that low dose irradiation had a ‘protective’ effect or induced an adaptive response in the context of a subsequent traumatic injury. Neurogenesis was performed to quantify numbers of newly born cells in the dentate gyrus. Bromodeoxyuridine (BrdU) was used to label all proliferating cells, and cell specific antibodies were used to indentify the specific types of cells that were produced. In terms of total numbers of newly born cells (BrdU+ only), independent of phenotype, there was a significant group difference, with more cells after trauma alone and less cells after irradiation alone, when compared to sham treated mice. After combined injury there was a trend towards higher numbers than seen following radiation alone. In contrast to the results for BrdU only, there was only a trend towards a group difference in newly born neurons, with trauma inducing a slight increase relative to sham treated mice, and irradiation causing a slight decrease relative to

Task Progress:	shams. After combined injury there was an increase in the numbers of newly born neurons compared to radiation alone. Taken together, these data did not appear to support our contention that a previous irradiation would induce a heightened vulnerability to a second insult. We also quantified newly born (BrdU+) cells associated with inflammatory processes: astrocytes and activated microglia. Astrocytes increased after trauma only, decreased after irradiation only and were not different from sham treated mice after combined injury. There were no differences between shams and animals that received trauma or irradiation only in terms of activated microglia. However there were more activated microglia after combined injury when compared to shams. Our observations relating to neurogenesis suggested that single or combined injury had differing effects on that process. However, the mere increase or decrease in numbers of newly born cells/neurons, or where they move, does not give any indication as to the functional status of those cells. To determine if newly born cells were functionally integrated into the hippocampal circuitry, we looked for the expression of Arc in BrdU+ neurons. Arc expression in newly born cells can be interpreted to mean that those cells are integrated, because in vivo, Arc is only expressed in neurons engaged in synaptic plasticity. Our data showed that the only mice that did not show any newly born-Arc expressing neurons were in the trauma only group; the group that showed significant cognitive changes. The significance of this finding needs further study. In our experimental paradigm, behaviorally-induced Arc protein was induced in mature granule cell neurons by the water-maze task. Trauma alone significantly reduced the expression of Arc protein relative to all the other treatment conditions in both hemispheres. This reduction was not due to changes in neuronal cell number. After combined injury there was an increase in the number of neurons expressing Arc protein relative to what was seen after trauma alone, suggesting that irradiation prevented the trauma-related decrease in Arc expression. The last study performed in this final year of the research project involved studies of cognition, Arc protein expression and neurogenesis in mice deficient in the gene for chemokine receptor-2 (CCR2). These mice have a muted inflammatory response relative to wild type mice, and we wanted to determine how this would affect our results after single and combined treatments. The only data available at this time are the cognitive studies; all other analyses are ongoing. Probe trial results showed that animals in all treatment groups showed a preference for the target quadrant indicating that mice had normal retention of spatial memory. This finding is in contrast to what we observed in wild type mice where there as a significant cognitive deficit after trauma alone. This suggests that a muted inflammatory response mediated through a loss of the CCR2 receptor plays a significant role after trauma, but not after irradiation or combined injury. More insight into this finding will hopefully become available once we have analysed the tissues for neurogenesis, Arc expression and numbers of activated microglia. The overall goal of this research proposal was to determine if a low dose of particulate irradiation induced changes in the brain that led to a heightened vulnerability upon a secondary traumatic brain injury. Unforeseen technical issues (i.e. anesthetic problems) had a dramatic impact on some of the early studies and precluded completion of some of the originally proposed experiments (e.g. early CCR2 and pharmacologic studies). However, during the course of the grant we were able to expand our goals to include molecular and behavioral endpoints that were not originally proposed. These inclusions greatly enhanced out ability to address factors critical to behavioral performance after single or combined injury. Overall, our combination of cellular (neurogenesis, neuroinflammation), molecular (Arc expression) and behavioral endpoints provided us quantitative means to show that a single 0.5 Gy dose of 56Fe ions affected measures of neurogenesis but not Arc expression or behavioral function. On the other hand, trauma alone significantly affected all the endpoints assessed to date. However, when irradiation preceded trauma by 30 days, there was no exacerbation of the effects of trauma, suggesting that under the paradigm used here, irradiation did not predispose the brain to a heightened vulnerability to a second injury. In fact, at least for our endpoints, it may be that a low dose of irradiation actual played a protective role. In a general sense, this resembles a preconditioning, adaptive or inducible-like radioprotective response where a sublethal or potentially injurious stimulus (i.e. irradiation) induces tolerance to a subsequent and potentially more damaging insult (trauma). The nature of the effect we observed and its mechanistic basis needs to be further studied. We originally proposed that inflammatory changes induced by irradiation may heighten the vulnerability to a secondary insult, but our data do not suggest this to be true. In fact our recent cognitive performance data in CCR2 seem to suggest that inflammatory processes only plays a role after trauma alone; however the analyses of cellular/molecular data are not yet complete. In conclusion, the data obtained during the performance of this grant proposal highlights a potentially important finding: that radiation may induce changes that modulate the extent of injury induced upon a second insult. These data also suggest that this may occur after a relatively low radiation dose (i.e. 0.5 Gy); one that is in line with what might be expected during a prolonged space mission. This could have important ramifications regarding potential strategies or approaches to the management of combined injury. Understanding the mechanism(s) responsible for the apparent 'preconditioning' effect could lead to potential pharmacologic interventions that could be used in the event of a post-irradiation traumatic event.
Bibliography Type:	Description: (Last Updated: 04/11/2018)
Articles in Peer-reviewed Journals	Rosi S, Belarbi K, Ferguson RA, Fishman K, Obenaus A, Raber J, Fike JR. "Trauma-induced alterations in cognition and Arc expression are reduced by previous exposure to 56Fe irradiation." Hippocampus. 2012 Mar;22(3):544-54. Epub 2010 Dec 29. http://dx.doi.org/10.1002/hipo.20920 ; PubMed PMID: 21192069 , Mar-2012